

Food waste reduction efforts in reducing greenhouse gas emissions and the impact of climate change: A review

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ABSTRACT

Food waste (FW) is a global challenge with profound environmental, economic, and social implications, particularly through its substantial contribution to greenhouse gas (GHG) emissions. This review aims to synthesise evidence on the relationship between FW reduction strategies and climate change mitigation, with a focus on its potential valorisation as bioenergy. A systematic examination of the recent literature was conducted, addressing the composition, physicochemical characteristics, and conversion technologies, including hydrothermal liquefaction, pyrolysis, fermentation, and anaerobic digestion. The findings indicate that FW accounts for up to 8 % of global GHG emissions, predominantly methane, which has a far greater warming potential than CO₂. Reduction and recovery strategies could mitigate millions of tons of CO₂-equivalent annually while generating high-value biofuels. Hydrothermal liquefaction emerges as the most suitable technology for high-moisture FW, with significant potential for yield improvement through pretreatment and catalytic enhancement. The novelty of this review lies in its integration of policy, practice, and technological perspectives into a unified analytical framework, encompassing regulatory instruments, supply chain innovations, and opportunities for developing a circular bioeconomy. In conclusion, reducing FW is not only an ethical imperative but also a practical pathway toward climate change mitigation, offering co-benefits for food security, energy sustainability, and intergenerational equity.

1. Introduction

Food waste (FW) has emerged as one of the most pressing global challenges, with substantial social, economic, and environmental consequences. A significant portion of food produced worldwide never reaches its intended consumers, ultimately decaying and being categorised as FW [1–3]. The steady increase in the global population has further intensified food consumption, thereby accelerating the accumulation of FW. This issue was exacerbated during the COVID-19 pandemic, which disrupted global supply chains and food processing

systems, resulting in even greater levels of food loss and waste.

The primary sources of FW are diverse, ranging from food processing industries, markets, restaurants, and kitchens to household-level waste. Within urban environments, FW has been reported to contribute up to one-third of municipal solid waste [4]. According to a United Nations (UN) report, an estimated 931 million metric tons (MT) of food are wasted globally each year, with households contributing 61 %, followed by the food service sector (26 %), and retail and distribution systems (13 %) [5]. In Canada alone, more than half of all FW originates from consumers (2.4 million tonnes), supermarkets (1.3 million tonnes), and

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industrial processing (2.8 million tonnes), a volume sufficient to feed the national population for five years [6].

Beyond economic losses, the environmental implications of FW are particularly severe. According to the U.S. Environmental Protection Agency (EPA), FW contributes approximately 170 million MT of CO₂-equivalent emissions annually [7–9]. When disposed of in landfills, FW decomposes anaerobically, producing methane (CH₄) and carbon dioxide (CO₂), both of which are potent greenhouse gases (GHGs) [10–12]. Globally, FW accounts for nearly 8 % of total GHG emissions, equivalent to 4.4 gigatons of CO₂ [13,14]. These emissions not only exacerbate global warming but also contribute to local environmental problems such as leachate generation and water pollution [4,15].

At the same time, FW presents both challenges and opportunities. The increasing volume of discarded food, driven by population growth and urbanisation, underscores the urgent need for sustainable management strategies. Properly managed, FW can be transformed into valuable resources such as compost and renewable energy. Such valorisation processes not only mitigate GHG emissions but also provide tangible economic benefits to society. Moreover, energy recovery from FW offers governments an alternative strategy to reduce dependency on fossil fuels while advancing renewable energy targets [13,14].

For this study, FW is defined as any edible or inedible material discarded at various stages of the food supply chain, including production, processing, distribution, and consumption, encompassing spoiled, unused, or otherwise unfit food for human consumption. Within this broad definition, the term leftovers refers explicitly to edible food portions that remain after meals are served or consumed at the household or food service level but are not reused or redistributed. To ensure clarity and consistency throughout this review, FW is used as the overarching concept, while leftovers are treated as a subset of FW, particularly originating from consumer and service-level practices [1,2,16,17].

The specific objective of this review is to synthesise and critically evaluate existing knowledge on FW management with a particular focus on its potential to mitigate climate change. Special attention is given to valorisation technologies such as anaerobic digestion, fermentation, pyrolysis, and hydrothermal liquefaction, which can convert FW into high-value products, including bioenergy, biofuels, and biochemicals. By integrating environmental, technological, and policy perspectives, this review aims to highlight the dual role of FW reduction: as a key strategy for mitigating greenhouse gas emissions and as a pathway to promote a sustainable circular bioeconomy.

2. Literature review

There are several benefits from FW valorisation, including environmental, social and economic, as reported in previous research [18–21]. FW valorisation has been widely introduced, including animal feed, biological fertiliser and chemicals [22–24]. However, as the population increases annually, the trend of increased FW continues to grow. The

trend of increased waste in Indonesia during the 2019–2022 period is shown in Fig. 1 [25]. Utilising FW in bioenergy in the form of fuel is one of the right ways to reduce dependence on fossil energy. FW, produced as fuel, can protect the environment, water bodies, and human health, while also maintaining national energy security.

Several methods can be applied to manage FW in bioenergy. The techniques that can be used in FW management can be based on the type of final product produced, such as solid, gas and liquid fuel, as presented in Fig. 2. Gas products produced from FW through torrefaction, anaerobic digestion, gasification, and combustion include carbon dioxide (CO₂), carbon monoxide (CO), bio-methane (CH₄), and hydrogen (H₂). The thermochemical conversion process through biomass gasification can produce temperatures of 600 °C to 1000 °C and low amounts of oxygen. Carbonaceous raw materials can be converted into gas with media (CO₂, H₂, air and steam) [26–29]. Torrefaction can generally be processed through pretreatment at 180–260 °C to remove water and volatile materials. In this way, the feed size can be the same to improve the fuel properties [30–32]. Energy production at temperatures of 500–1000 °C can be achieved through the thermochemical conversion of biomass, which is more suitable for use with forest waste. Bicarbonate and carbonate compounds, such as CaCO₃, are generally produced from combustion at temperatures below 500 °C and contain a lot of ash.

Meanwhile, burning at 1000 °C produces oxide compounds, such as whiting (CaO). Additionally, light metal elements such as sodium,

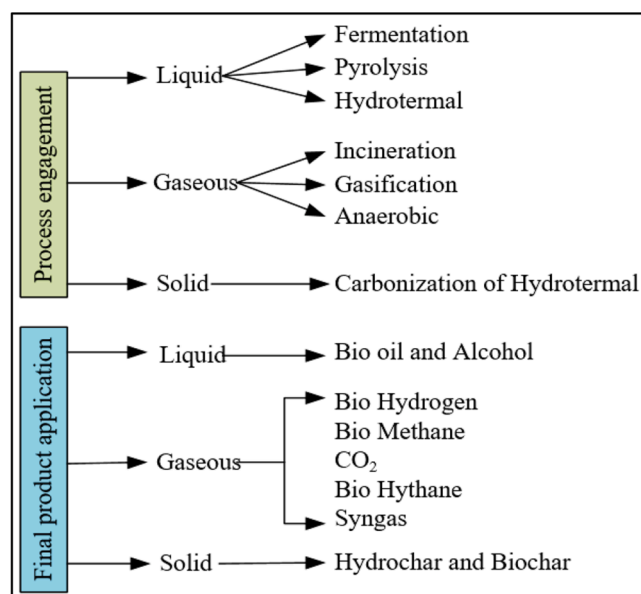


Fig. 2. FW processing methods according to type and product.

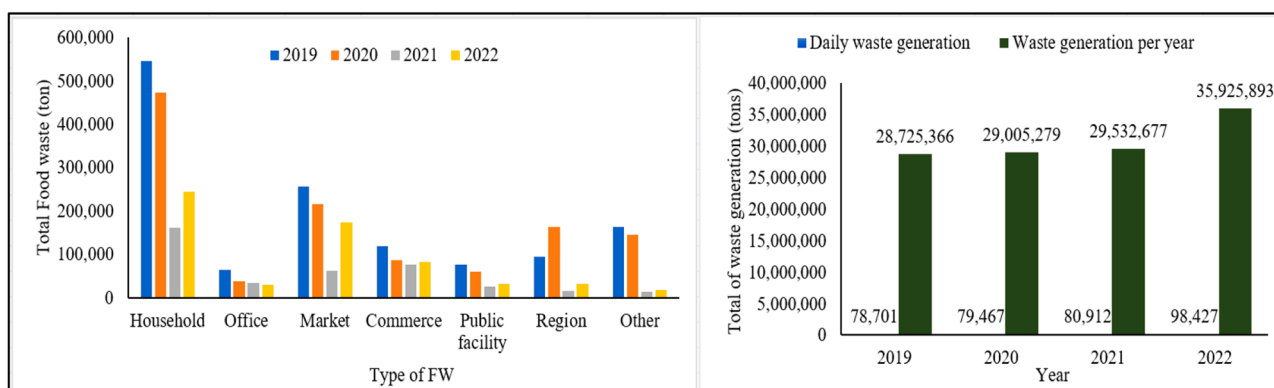


Fig. 1. Data on the composition and generation of waste in Indonesia.

potassium, and zinc can reduce ash as the combustion temperature increases [33–35]. Incineration technology has been considered an appropriate method for waste management [36–38]. Meanwhile, for FW with high water content, anaerobic digestion technology can be used to produce maximum energy [39–41]. Aerobic, hydrolytic reactions and fermentative microorganisms can convert FW into carboxylic acids, especially CO₂, hydrogen, acetic acid, and alcohol. The carbon groups and aromatic rings produced from FW HTL reach 26.4–34.4 % [42–44].

Biocrude is one of the final products obtained from hydrothermal liquefaction, fermentation, and pyrolysis. Pyrolysis and hydrothermal liquefaction are the most widely used methods for processing FW-based bio-crude. The thermochemical process through hydrothermal liquefaction can be pressurised at 5–25 MPa and a temperature of 200–380 °C, which is used to process dry and wet waste, converting it into bio-energy such as hydrochar, gas, biocrude, and a liquid phase, in the presence of environmental oxygen [45–48]. Meanwhile, the thermochemical process in an oxygen-free environment can be carried out using the pyrolysis method, allowing biomass to undergo thermal degradation at temperatures exceeding 1000 °C [49–51]. The production of FW-based alcohol biofuel through fermentation using *Saccharomyces cerevisiae* has previously been carried out [52]. Additionally, producing hydrocarbons through carbonisation for final applications is a thermochemical process that yields solid, condensed hydrocarbons based on biomass waste [53–55]. Different studies report that the water content of industrial FW reaches 53 % and contains grain products. Apart from that, FW from industry also offers fresh vegetables, processed fruit, milk, fish products, chicken, meat, nuts, eggs, and sweeteners, which are processed by hydrocarbon carbonisation and hydrothermal melting at temperatures ranging from 190 °C to 340 °C. Their research findings indicate that low liquid yields were obtained at high temperatures, accompanied by increased gas production and a reduction in biochar production. However, when the lipids are hydrolysed at 202 °C, the biochar appears carbonised, but it rapidly accelerates when the temperature is ≥ 250 °C [56]. Thus, low-temperature hydrothermal liquefaction can reduce energy costs but can complement lipid hydrolysis to produce long-chain fatty acids.

The comparative analysis of food waste valorisation pathways in Table 1 highlights clear distinctions among anaerobic digestion (AD), pyrolysis, and hydrothermal liquefaction (HTL) with respect to operating conditions, yields, suitability, and technological maturity. AD operates under mild mesophilic to thermophilic conditions (35–55 °C) with relatively simple pretreatment, producing biogas and digestate, making it highly suitable for wet and heterogeneous food waste streams. However, despite its positive energy return on investment (EROI ≈ 3 –5), the low energy density of methane gas and purification requirements remain major drawbacks. Pyrolysis, in contrast, requires high-

temperature inert conditions (300–1000 °C) and strict feedstock preparation (drying, low moisture), producing a spectrum of products including bio-oil, biochar, and syngas. While yields are promising (bio-oil 15–30 wt%, biochar 25–35 wt%, gas 20–40 wt%), the moderate heating value (16–20 MJ/kg) and instability of bio-oil necessitate upgrading, limiting large-scale deployment to pilot or semi-commercial levels. HTL emerges as a strong candidate for high-moisture food waste valorisation, as it employs water as a reactive medium under subcritical conditions (200–380 °C, 5–25 MPa) without requiring drying. It achieves high biocrude yields (30–60 wt%, exceeding 70 % with catalysts) with superior heating values (30–47 MJ/kg), approaching the quality of fossil crude after upgrading. Nevertheless, HTL still faces challenges such as high-pressure operation, costly equipment, and corrosion issues, which constrain its readiness level to lab and pilot scales. Overall, the synthesis illustrates that while AD is already commercially established, pyrolysis and HTL offer broader product portfolios and higher-quality fuels, with HTL showing the most significant long-term potential for sustainable food waste valorisation once technical bottlenecks are resolved.

Organic waste and FW have great potential to fuel lipid synthesis through fungi and yeast because they contain oil [78–80]. Low-cost carbon has now been widely used for primary raw materials, including FW, refinery waste, municipal wastewater, rapeseed brine, and hydrolysis of agricultural waste [81–85]. Due to the lower concentrations of hazardous substances and heavy metals, wastewater from the production of drinks and food is more suitable for cultivating microalgae. Meanwhile, potassium, lipids, iron, saccharides, phosphates, minerals, and organic acids are lower in waste from soybean processing, making the substrate more suitable for microalgae cultivation [68,86,87]. Lipids found in yeast cells include *Lipomyces* sp., *Rhodotorula* sp., and *Rhodospiridium* sp. Equivalent to 70 % of dry biomass [88]. Fermentation using wastewater sludge for FW can increase hydrogen output by more than 14 times [89,90]. Biomass pollution has a lower level of cooling complexity with a volatile solid's ratio VS/TS > 0.7 , with a nutrient balance ratio C/N 20–30 and N/P 6.6–7. Microbial activity and biomethane production increase with these properties in biomass. This is due to factors such as FW's energy production process and the increase in anaerobic pollution compared to other forms of biomass [68,69,91].

In general, the manufacture of bioalcohol currently uses raw materials from food plants, which are the first generation. The investigation of alternative, low-cost raw materials is underway, as food security worldwide is facing a crisis compared to traditional terrestrial raw materials. Microalgae, with their good photosynthetic activity, are a promising option because they can grow in water. Additionally, it possesses a favourable chemical composition and structural features,

Table 1
Comparative of food waste valorisation pathways.

Parameter	Anaerobic Digestion (AD)	Pyrolysis	Hydrothermal Liquefaction (HTL)
Main products	Biogas (CH ₄ 50–70 %), digestate [39–41].	Bio-oil, biochar, syngas [49–51].	Biocrude, gas, aqueous phase, hydrochar [45–48].
Operating conditions	Mesophilic/thermophilic 35–55 °C, atmospheric, retention 15–30 days [39–41].	300–1000 °C, inert atmosphere, seconds–minutes [57–61].	200–380 °C, 5–25 MPa, 30–60 min [57–61].
Pretreatment needs	Low (basic mixing/homogenisation) [22].	High (requires drying, moisture < 10 %) [22, 23].	Drying not required; optional pretreatment (enzymes, catalysts, ultrasonication) [62–64].
Suitability for high-moisture FW	Very suitable [39–41].	Not suitable (dry feedstock required) [65,66].	Highly suitable (water acts as a medium) [65–67].
Typical yields	0.3–0.5 m ³ CH ₄ /kg VS (≈ 20 –25 MJ/kg VS) [68, 69].	Bio-oil 15–30 wt%, biochar 25–35 wt%, gas 20–40 wt% [49–51].	Biocrude 30–60 wt% (can exceed 70 % with catalysts), HHV 17–47 MJ/kg [70–72].
Energy efficiency/balance	Positive, EROI ≈ 3 –5; but low energy density (gas) [68,69].	Moderate; HHV of bio-oil 16–20 MJ/kg, unstable, needs upgrading [49–51].	High; HHV of biocrude 30–47 MJ/kg, approaching fossil crude after upgrading [70–72].
TRL (Technology Readiness Level)	High (commercial, worldwide deployment) [39–41].	Medium (pilot/semi-commercial, oil upgrading limits) [57–61].	Medium–low (lab/pilot scale, limited industry application) [45,73,74].
Advantages	Mature technology, which directly utilises CH ₄ , mitigates landfill emissions [39–41].	Produces multiple products (bio-oil, char, syngas), potential co-firing [49–51].	Efficient for wet FW, biocrude similar to petroleum [45–48].
Limitations	Produces low-energy-density gas, requiring purification [39–41].	Requires drying, unstable oil, costly upgrading [22,23,75,76].	Requires high pressure, costly equipment, and corrosion issues [22,23,74,77].

making it a promising candidate for microbial fermentation and bioalcohol synthesis, particularly in the context of biorefinery. Raw materials for biodiesel production are FW through a microbial cell factory process. Biochemical and thermochemical conversions using algae can produce biopolymers, biofuels, and bioproducts, and can be scaled up to an industrial scale [92,93]. Biomass and microalgae fractionation are two examples that can be used to solve the trilemma [94–96]. The contribution of microalgae plays a significant role in connecting food production, water purification, air purification, and the production of raw energy materials. Microalgae are a new way to process FW because they can help achieve a circular economy. Conversion of biomethane from FW can be done through anaerobic digestion. A source of nutrients is necessary for microalgae growth, allowing them to be utilised in recycling nutrients for freshwater digestion [97–99]. Gradually, microalgae have been used and researched in reallocating FW nutrients to be used and utilised for energy and fertiliser products so that a circular economy can be achieved [100–102]. Investigating FW raw materials with a ratio of mixed vegetables and turkey, yellow oil, and algae can increase the asphaltene content and lower the biocrude [103]. The research results reported that adding yellow oil to FW could increase biocrude by 24 % and 39 %, respectively, with mixtures of 10 % and 25 % yellow oil. The asphaltene content in biocrude decreased by 26 % without yellow oil, 48 % with 10 % yellow oil, and 85 % with 50 % yellow oil.

Of the methods described above, hydrothermal liquefaction is the most suitable method for FW, where there is a lot of water vapour. Processing FW with hydrothermal liquefaction has attracted researchers worldwide because it uses water naturally and does not require initial drying for pyrolysis. Thus, the process can save energy, as reported in previous studies [66,67,73,107]. Meanwhile, the advantages and disadvantages of hydrothermal liquefaction and pyrolysis methods are summarised in Table 2.

An overview of the number of publications, including review articles, research studies, and books, as well as various other publications related to fuel conversion from FW, especially those published in Elsevier, is presented in Fig. 3(a). Research trends employed in the field of FW over the last decade are closely tied to the topic, with demand expected to continue increasing in the future. Apart from that, publications related to the conversion of FW into fuel from agricultural waste have also

become popular in recent years, as shown in Fig. 3(b). Meanwhile, the potential of various FW being converted and producing fuel is also a topic of interest among researchers, as presented in Fig. 4. Meanwhile, the types of FW that are suitable for biocrude production raw materials are also discussed in this work.

A poor storage system can spoil food over time, resulting in unusable FW. However, if the food storage system is in the correct and appropriate conditions, it will not be damaged and can be used for a long time. If the moisture in the FW has been lost, it does not affect the hydrothermal liquefaction process because it can be added during processing. However, on the contrary, the water content in food increases due to spoilage, which in turn increases the weight, thereby decreasing the yield of biocrude production. In addition, microbial growth is facilitated by a decrease in FW, allowing the lignocellulosic biomass to be consumed. The cellulose, hemicellulose and lignin content in biomass changes, which can affect the final biocrude process [48,108,109]. Thus, fresh raw materials require storage conditions and should be used immediately. As the research results show, FW slurry is stored at a temperature of 4 °C before being used for hydrothermal liquefaction for 2–3 days [46–48,110].

Currently, viable biofuel converted from FW on a pilot scale is unavailable in various countries because research is generally carried out on an autoclave scale. Apart from that, FW pre-treatment to increase biocrude yield through hydrothermal liquefaction has not yet been widely explored. Therefore, this review discusses the knowledge gaps regarding FW-based biochemical properties in comparison to feasible processing systems. This work broadly addressed the composition of FW and its proximate analysis, including elements that can be used to produce biofuel through hydrothermal liquefaction. This work also discusses the FW pre-treatment method for building high biocrude and explores various side applications in hydrothermal liquefaction. This research also summarised and analysed the final process of biocrude through hydrothermal liquefaction made from FW raw materials and unnecessary raw materials, mixing, upgrading, and co-processing. Previous developments in hydrothermal liquefaction reactors and their economics are also essential to note. Bibliometric analysis with joint network presentation in research on hydrothermal liquefaction of FW in recent years is shown in Table 3.

3. Methodology

This study employed a systematic literature review approach to synthesise and evaluate recent developments on food waste (FW) management, its contribution to greenhouse gas (GHG) emissions, and the potential for valorisation into bioenergy. The methodology was designed to ensure transparency, reproducibility, and reliability of the findings.

3.1. Literature search strategy

Relevant publications were identified from major academic databases, including ScienceDirect, Scopus, Web of Science, and Google Scholar. The search covered the period from 2013 to 2023, reflecting the most recent decade of research on FW valorisation. Keywords used in combination included food waste, greenhouse gas emissions, climate change mitigation, anaerobic digestion, fermentation, pyrolysis, and hydrothermal liquefaction. Additional references were retrieved from the bibliographies of selected papers.

3.2. Inclusion and exclusion criteria

The inclusion criteria for this review encompassed peer-reviewed journal articles, review papers, conference proceedings, and policy reports directly related to FW management and valorisation. Only studies providing quantitative or qualitative data on FW composition, physicochemical characteristics, conversion technologies, or their

Table 2
Pyrolysis vs. hydrothermal liquefaction.

Parameter	Pyrolysis	Hydrothermal liquefaction	Ref.
Easier process	Requires a pre-drying process	Easier	[104]
The resulting bio-oil has fixed parameters.	The least	Additional	[105, 106]
Promising economy	Heat treatment makes the process more expensive	Low cost	[22,23]
Pre-treatment Condition and type	Required Low temperature pyrolysis 300–700 °C, 0.1–10 °C/minute High temperature pyrolysis 60–1000 °C, 10–1000 °C/minute and 900–1300 °C, >1000 °C/s	Not required Temperature 200–350 °C, Pressure 10–15 MPa, and Time 30–60 s	[22,23] [57–61]
Suitability for FW	Before the reaction process, the FW must be dried first.	Accordingly, water in food products can be used in reactors.	[66,67, 104]
Save energy usage	Poor	Better	[66,67]
Higher HHV, MJ/kg	Less	Higher	[71,72]

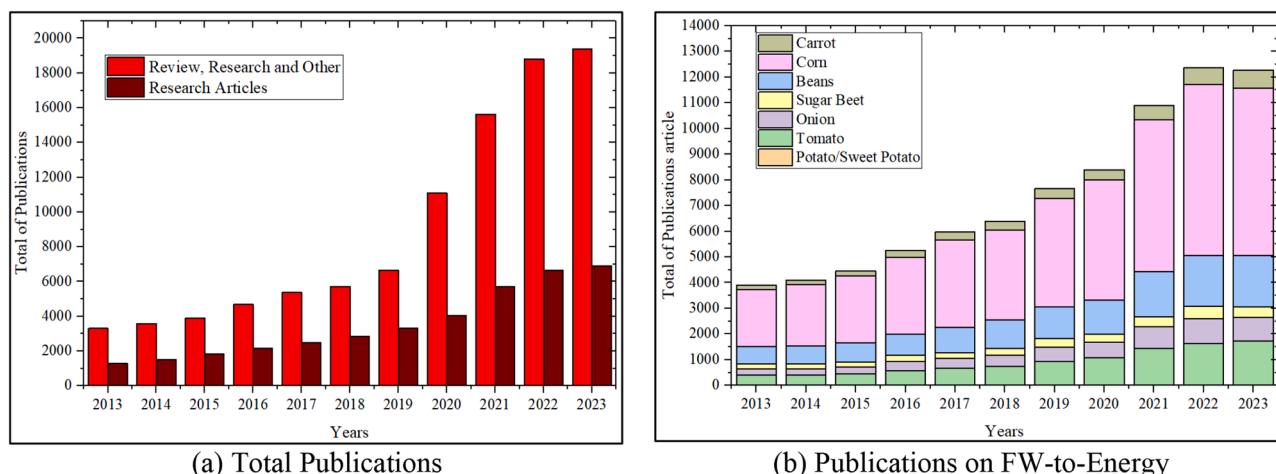


Fig. 3. Publications on agricultural food waste (FW) valorisation (2013–2023): (a) total publications in the Elsevier database, (b) publications on FW-to-energy conversion by crop type.



Fig. 4. Types of FW that can potentially be biofuel.

environmental impact were considered relevant. Publications in English and Indonesian were included, provided that they explicitly discussed the link between FW and GHG emissions or renewable energy recovery. In contrast, papers unrelated to FW or lacking environmental or energy perspectives were excluded, as were studies without methodological transparency or empirical evidence. Grey literature that had not

undergone peer review was generally excluded unless it contributed essential policy information.

3.3. Data extraction and analysis

Data from the selected studies were extracted and analysed through

Table 3
Bibliometric analysis in research on hydrothermal liquefaction of FW.

Parameter	Final product	Type of FW	Countries	Ref.
Enhancement of biocrude quality using hydrodenitrogenation	Biocrude	Organic waste from municipal food sources	Italy	[111]
Environmentally friendly diesel generators are promising for the future prosperity of the biofuel industry.	Biofuels such as bioethanol, biodiesel, biogas, bio-oil, and green diesel	Agricultural wastes	Malaysia	[112]
Evaluations of the biocrude's production and quality	Biocrude	FW for Municipal	India	[113]
Recovering nitrogen and improving the quality of biocrude	Biocrude	leftovers from the processing of food	China	[114]
The implementation of Hydrothermal melting in practice	Biocrude	Expendables from the manufacturing of tobacco	Thailand	[115]
Items with added value made from organic waste	Biochar	Compostable material	India	[116]
The movement of components throughout the HTL process of the feedstock	Biocrude	FW and Other	China	[117]
Accuracy of the learning process used by machine-learning models to forecast biocrude yield	Biocrude	FW	US	[118]
The influence of a variety of solvents and catalysts on the hydrothermal gasification and liquefaction processes	Crude biofuel and crude biohydrogen	FW for Household	India	[119]
The quantity and quality of biocrude	Biocrude	Compostable material	Italy	[120]
Observations on mixed feedstock behaviour in Hydrothermal melting	Biocrude	Expendable foods, grass clippings, yard trimmings and wood chips	US	[121]
Emulsification processes, such as mixing time, temperature, stirring intensity, and emulsifier	Biocrude	Waste wheat flour and canola flour	India	[122]
A high heating value, total acid number, amount of oxygen and nitrogen, and viscosity of biocrude	Diesel fuel	Plant for the processing of local foods	US	[123]
Recent developments around thermochemical transformation	Biocrude	Waste from the kitchen	China	[124]
Analysis on a larger scale of the production of biocrude and its upgrading	Biocrude	FW	India	[125]
Analyses of the economy	Biogas	FW for Landfill	US	[126]
Combining characteristics	Biocrude	Chicken and various vegetables	Canada	[103]

three complementary steps. First, publications were categorised thematically into clusters focusing on (i) FW and its contribution to greenhouse gas emissions, (ii) conversion technologies and valorisation pathways such as anaerobic digestion, fermentation, pyrolysis, and hydrothermal liquefaction, and (iii) policy, social, and economic perspectives on FW management. Second, a bibliometric analysis was conducted to identify trends in publication outputs and highlight the most frequently studied pathways from FW to energy. This was complemented by graphical and tabular summaries in the manuscript, including figures that illustrate publication dynamics and tables that synthesise bibliometric findings, particularly for hydrothermal liquefaction. Third, a comparative synthesis was performed by examining selected studies in terms of their operational conditions, energy efficiency, product yields, and technology readiness levels, thereby enabling a structured comparison between different FW conversion pathways.

3.4. Validation and limitations

To strengthen the validity of the review, the analysis incorporated evidence from multiple data sources, including experimental studies, review articles, and bibliometric datasets, thus ensuring a robust triangulation of information. Nevertheless, several limitations must be acknowledged. The scope of the review was inherently influenced by database coverage and the restriction to English and Indonesian publications, which may have excluded relevant studies in other languages. Moreover, the heterogeneity of reporting methods, experimental setups, and performance metrics across the reviewed studies limited direct quantitative comparisons. Finally, the scarcity of large-scale pilot or industrial data, particularly for hydrothermal liquefaction, restricts the generalisability of certain conclusions. Despite these constraints, the methodological approach ensures that the review provides a representative overview of the current state of knowledge, while highlighting research gaps and future directions for the sustainable management of food waste.

4. Analysis of the quality of food waste

4.1. Chemical composition of food waste

The type and quality of suitable biofuel depend on the FW composition and the quality of the food source itself. FW has biodegradable properties, making it an excellent candidate for use as a biofuel due to its high organic content. Classification of various FW according to the US Department of Agriculture, as presented in Fig. 5 [127,128].

4.2. Analysis of food waste

Composition, source, and constituents will significantly influence the physical-chemical properties of FW, including density, water content, and pH [4,15,129]. FW containing high water content will increase the size of particles and aggregates, increase heat and reduce oxygen [130,131]. Additionally, volume and porosity decrease due to the higher water content [132,133]. The concentration of hydrogen ions in FW generally exhibits acidic properties, as it is composed of components that facilitate the bioconversion process [134–136]. Meanwhile, the bulk density of FW typically reaches 480 kg/m³ and has high solids and low porosity [137,138]. The complete approximate analysis of the biochemical composition of various FW is presented in Table 4.

There are three main polymers in FW, each containing 15–40 % lipids, 30–60 % carbohydrates, and 5–20 % protein, which vary significantly according to the composition and type of waste [152,153]. The complete biopolymers contained in FW are presented in Table 4. The cereals contained in FW contribute significantly to carbohydrates. Likewise, meat products, nuts, and eggs can contribute to protein and lipid content. The elevated lipid content in FW will produce more liquid biofuel [154–156]. The comparison results of pure FW with mixed FW via hydrothermal liquefaction also show the same thing [46]. The reported results of biocrude are 78 % higher than those of salad dressings containing lipids. Carbon yield reached 88 %, and energy recovery reached 98 % compared to FW mixture collection, which only achieved 47 % biocrude.

An overview of the combustion properties of FW can be obtained from the proximate analysis results, including fixed carbon content,

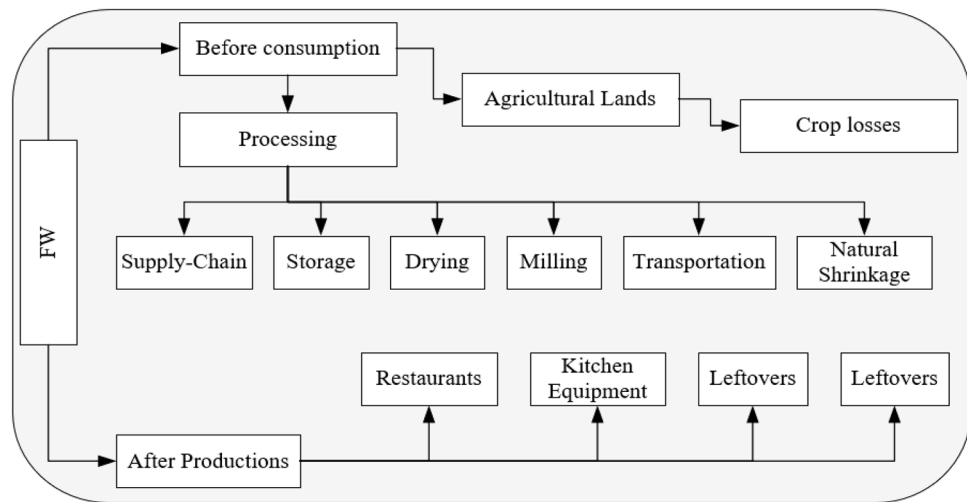


Fig. 5. Classification according to type of FW.

Table 4
The biochemical makeup and proximate analysis of various types of FW.

Type of FW	HHV (MJ/kg)	Volatile Matter	Moisture Content	Ash	pH	Carbohydrate	Lipid	Ref.
Halls of dining	24.5	NA	70.13 %	4.7 %	NA	NA	NA	[46]
Cafeteria Food	NA	NA	75.9 %	1.3 %	NA	71.6 %	NA	[139]
Vegetables	22.5	NA	85.2 %	3.2 %	NA	49.5 %	15.4 %	[67]
Restaurants	NA	NA	80 %	4.9 %	NA	53.1 %	20 %	[140]
Halls of dining	24.6	94.6 %	86.7 %	NA	NA	NA	NA	[45]
Rice, meats, and vegetables for restaurants	NA	98.2 %	81.9 %	NA	NA	90.6 %	NA	[141]
Restaurants for students	NA	94.1 %	81.5 %	5.9 %	NA	95.9 %	14 %	[142]
Palm oil mill waste	NA	NA	70.5 %	1.6 %	4.5	NA	3.8 %	[143]
Bakery	NA	NA	22.3–45 %	NA	NA	35.5–62 %	0.5–32.5 %	[144]
The kitchen of the household	NA	95.4 %	80.3 %	1.9 %	NA	61.4 %	15.7 %	[145]
FW from a factory salad dressing plant	NA	NA	80–99 %	6.2 %	NA	50.3 %	40.8 %	[146]
Garbage from kitchens	NA	98.2 %	85.7 %	NA	NA	70.6 %	NA	[147]
FW of landfill	NA	15.3 %	62.2 %	5.4 %	NA	NA	NA	[148]
FW from kitchen scraps and cafeterias	NA	NA	80 %	5.8 %	NA	53.6 %	18.6 %	[140]
Garbage from kitchens	NA	89.1 %	82.8 %	NA	3.4–6.6 %	98.2 %	18.1 %	[147]
FW from retail stores	NA	NA	64 %	NA	NA	63.5 %	4.1 %	[149]
FW from the household kitchen and cafeterias	NA	NA	61.3 %	1.2 %	NA	69 %	6.4 %	[150]
FW grains, meats, and vegetables from cafeterias	19.5	18.7 %	79 %	2.3 %	NA	NA	NA	[151]
FW from Restaurants	NA	87.5 %	81.7 %	NA	NA	35.5 %	24.1 %	

water content, ash content, volatile substances, and others [157–159]. The remaining results from burning FW are referred to as ash content, which can contribute to the properties of charcoal. The heating value of FW does not increase in the presence of high ash content. Meanwhile, volatile materials in FW will produce liquid/gas fuel during combustion [4]. In general, the proximate analysis obtained from FW based on previous research is presented in Table 4.

Additionally, an analysis of the chemical properties of FW was obtained from the ultimate analysis. The leading analyses studied are generally carbon, hydrogen, nitrogen, oxygen, and sulfur. These results are needed to balance the mass and C/N according to the FW [152,153]. The results of the ultimate analysis, obtained from several previous publications, are presented in Table 5.

After discussing the contribution of FW to greenhouse gas emissions, it is crucial to outline the pathways available to mitigate these impacts. FW management involves a series of strategies that transform waste into valuable resources while reducing environmental burdens. These strategies typically begin with pre-treatment processes that prepare FW for conversion, followed by a range of valorisation technologies such as anaerobic digestion, fermentation, and thermochemical methods. The end products include biofuels, bioplastics, organic acids, and other bio-based materials that support a circular economy. The following sections present these technical approaches in detail, starting with pre-treatment

methods and progressing through different conversion technologies and their applications.

To address the challenges posed by FW and its contribution to greenhouse gas emissions, the following sections provide a structured discussion of management strategies. The review begins with pre-treatment methods, which prepare FW for further processing. It then examines conversion technologies, including anaerobic digestion, fermentation, and thermochemical approaches, that transform FW into usable forms. The discussion continues with valorisation pathways, highlighting the production of biofuels and other bio-based products that contribute to a circular economy. Finally, the review examines policy frameworks and implementation aspects, which are crucial for ensuring the adoption and scalability of FW management strategies. This logical flow guides the reader from the problem of emissions to practical solutions and their broader implications.

Fig. 6 illustrates the holistic flow from the primary issue of food waste and its contribution to greenhouse gas emissions, towards mitigation opportunities through bioenergy conversion and circular economy strategies. The framework highlights the significant impacts, outlines key objectives, and presents the processing stages, including anaerobic digestion, fermentation, pyrolysis, and hydrothermal liquefaction. It further connects these technologies to final products, such as renewable energy, fertilisers, bioplastics, and chemicals, while also

Table 5
Ultimate analysis of food waste.

Type of FW	Rasio C/H	C	H	O	N	Ash	Ref.
Halls of dining	7 %	56.2 %	8.1 %	33.2 %	2.6 %	NA	[46]
FW From landfill	7.3 %	45.3 %	6.2 %	46.5 %	2 %	NA	[148]
Discarded products of cheese production	7.2 %	60.7 %	8.5 %	26 %	3.3 %	1.5 %	[160]
FW from orange peels	8 %	49 %	6.1 %	40.4 %	4 %	2.2 %	
FW from cafeterias and kitchen scraps	6.8 %	49.3 %	7.3 %	35.5 %	3.5 %	6.5 %	[140]
FW from restaurants	6.8 %	51.5 %	7.6 %	34.3 %	3.3 %	4.1 %	
FW from palm kernel shell	11.8 %	47.8 %	4.1 %	47.6 %	0.5 %	NA	[73]
FW from empty fruit bunches	10.8 %	43.6 %	4 %	50.2 %	2 %	NA	
FW from palm mesocarp fibre	9.9 %	46.3 %	4.7 %	47.4 %	1.4 %	NA	
FW from yoghurt whey	7.6 %	19.8 %	2.6 %	21.9 %	0.7 %	54.1 %	[161]
FW from the oil pomace of olive	7.9 %	56.1 %	7.1 %	31.5 %	2.5 %	3.9 %	
FW from miscellaneous	NA	47.9 %	NA	NA	2.8 %	NA	[162]
FW from salad processing	6.8 %	54 %	7.9 %	26 %	0.6 %	7.5 %	[160]
FW from fish meat	6.8 %	58.5 %	8.6 %	21.9 %	11.1 %	0.01 %	[163]
FW from whiskey stillage	7.2 %	46.9 %	6.5 %	38.8 %	4.1 %	3.7 %	[161]
FW from apple pomace	7.3 %	48.1 %	6.6 %	42.6 %	1.1 %	1.7 %	
FW from vegetables	6.8 %	49.3 %	7.3 %	37.4 %	5.8 %	NA	[67]

emphasising policy and practical aspects, including reduction targets, food labelling, and donation programs. Overall, the diagram encapsulates the integrated approach of food waste reduction, valorisation, and sustainable development.

4.3. Properties of the physical realm

The physical properties of food waste play a crucial role in the bioconversion process, which in turn affects its effectiveness and efficiency. Food Waste has several physical properties, such as moisture content, bulk density, colour, odour, particle size, and magnetic properties, which all play a role in the various bioconversion processes [130, 132,164]. Due to their role in the bioconversion of food waste, two key physical parameters — namely, moisture content and bulk density — are examined further below.

The amount of water in food waste is referred to as its moisture content. Alternatively, this may be thought of as the amount of water lost during drying when the food waste is heated to 105°. Food waste usually has a moisture content ranging from 50 % to 80 % water based on its dry weight. When a higher percentage of moisture is present, dry-weight food waste will have a higher value. A few researchers have suggested that increasing the amount of water that food waste contains will be beneficial in forming particle aggregates and increasing their size [130]. However, a decrease in porosity and an overall trend towards a lower pH can be observed when the moisture content of food waste is increased [132]. The unit of bulk density (BD) is represented as kilograms per cubic metre kg/m³. The total density of food waste, BD, can be defined as mass per unit volume [164]. The BD value is the most important indicator of both the porosity and the compaction of waste

material. The food waste typically has a BD ranging from 120 to 480 kg/m³ [165]. A high BD value indicates a minor porosity and substantial food waste compaction in the material.

4.4. Proximate and ultimate analysis

The percentages of moisture, volatile matter, ash, and fixed carbon that comprise food waste are determined through proximate analysis [157]. Proximate analysis is also used to determine the combustibility characteristics of the trash. The actual weight of trash is increased because of its high moisture and ash content, but this does not affect its heating value. During the combustion of waste, a portion of the trash is converted into a gaseous or liquid fuel, known as volatile matter. Fixed carbon is the small quantity of residue left over after completely burning garbage; it possesses the same quality as charcoal. The approximate analysis and ultimate analysis of food waste are presented in Table 6.

There are five main elements: carbon (C), hydrogen (H), nitrogen (N), oxygen (O), and sulphur (S). The ultimate analysis measures the amount of each one. The elementary analysis is another name for ultimate analysis [152]. The results of this study will help mix trash properly, ensuring it has the correct C/N ratio and mass balance for a chemical or thermal process. For bioconversion of food waste to work well, the C/N ratio should be between 10 and 30.

4.5. Pre-treatment of food waste

Hydrothermal liquefaction generally does not require pretreatment; however, some pretreatment is necessary to enhance biodegradation and facilitate hydrogen reactions. The essential treatment is grinding until it becomes denser, as FW contains higher levels of organic material [22]. However, hydrothermal pretreatment can also be done to increase biocrude production yields. It can also be done via microwaves, fungi, enzymes, thermal, alkaline, ultrasonication, bead grinding, and high-pressure homogenisers [4,62,63]. Initial treatment with bead grinding was performed using a mild process with 65 % ZrO₂ beads. This is done to select proteins rather than carbohydrates against microalgae, *Tetraselmis suecica*, and *Perchloromonas oleovorans* [64,176,177].

Meanwhile, a microwave treats *Chlorella sorokiniana* and *Bortyococcus* sp. initially at 100 °C for 2–6 min at 2450 MHz. This treatment can increase lipid extraction and palmitate content by 34.20 % [178]. Although pretreatment has been studied on microalgae, pretreatment in FW processing is still very rarely carried out, so it is imperative to explore it further. The initial treatment of microalgae has an adverse effect, especially on the results of biocrude production during processing at high temperatures. However, when processed at low temperatures, the initial treatment showed positive results [179,180]. Changes in biocrude production yield with pretreatment during hydrothermal liquefaction are summarised in Table 7.

4.6. Hydrothermal melting

The high-water content of up to 70–90 % in both wet and dry raw materials can be reduced through hydrothermal liquefaction. Generally, natural materials with a high-water content are obtained from FW. The hydrothermal liquefaction method is the cheapest for processing FW. The principle of this technology is to heat FW at 374 °C under a pressure of 22 MPa, allowing for the production of biocrude at both low and high temperatures [48,182]. During hydrothermal liquefaction at temperatures ranging from 200 °C to 380 °C and pressures of 5–25 MPa, the dielectric constant is reduced, enabling organic materials to mix with water. Thus, such conditions can speed up the reaction due to increased ionic products, making the gas homogeneous. In addition, this reaction also allows single-phase reactions to occur through resistance to mass transfer [66,67,183].

To operate hydrothermal liquefaction sustainably, it is recommended to use homogeneous and stable raw materials with particle sizes



Fig. 6. Conceptual framework of food waste valorisation pathways.

smaller than 100 μm [184,185]. Depolymerisation and hydrolysis occur due to the presence of macromolecules, so that the molecules that break down become minor, such as phenols, monosaccharides, fatty acids, amino acids, etc. [186–188]. This product undergoes various reactions at elevated temperatures in the reactor, including dehydration, decarboxylation, decomposition, and deamination [71,189,190]. At a temperature of 180 $^{\circ}\text{C}$, monosaccharides decompose first; when the temperature reaches 340 $^{\circ}\text{C}$, other molecules also deteriorate. In this case, the lipids and proteins contained in FW are of considerable significance [191].

Meanwhile, proteins can be hydrolysed when the temperature reaches 250 $^{\circ}\text{C}$ and degraded into amines, acids, hydrocarbons, and aldehydes [192–194]. Lipids are the highest biocrude yield from the hydrothermal liquefaction process, followed by proteins and carbohydrates [195,196]. Raw materials high in carbohydrates require reactions at high temperatures to be degraded. Thus, two main parameters can influence the biocrude product, including residence time and reaction temperature. Longer residence times and temperatures above 350 $^{\circ}\text{C}$ can lead to polymerisation reactions and biocrude decomposition, resulting

in more gas and char [197]. The quality of biocrude and yield through the hydrothermal liquefaction is further improved by a mixture of several catalysts. In general, heterogeneous catalysts are more popular because they are more economical than homogeneous catalysts. This is because homogeneous catalysts require an extraction process for recycling and separation [67,198]. They have increased heating value and higher biocrude by adding several types of catalysts from various FW, as summarised in Table 8.

Efforts to produce biocrude with various types of FW carried out by several researchers have shown promising results. The enzymatic hydrolysis process, using a 2:1:1 ratio of protease, lipase, and carbohydrase, before the hydrothermal liquefaction process, can increase biocrude production, with HHV reaching 17.4–26.9 MJ/kg and carbon by 43.7–65.4 % [70]. The lowest HHV recorded from carrot waste was 15.2–15.6 MJ/kg. The low HHV of carrot waste was attributed to its high water content, which ranged from 86.9 % to 89.4 % [206]. Therefore, FW from carrots shows low biocrude yield compared to other FW. However, the hydrothermal liquefaction method is more promising because the reaction medium used is water, allowing FW, which

Table 6

Proximate and ultimate analysis of food waste.

Ref.	Analysis of Proximate					HHV (MJ/kg)
	Volatile Matter	Moisture Content	Fixed Carbon	Ash	Type of FW	
[152]	NA	74 %	24.85 %	1.15 %	FW	22.7
[166]	93.55	80.03 %	NA	1.27 %	FW	NA
[167]	80.9 %	60.9 %	11.4 %	1.61 %	FW	NA
[157]	79.1 %	76.5 %	14.5 %	6.4 %	FW	15
[168–170]	18–30 %	60–85 %	NA	5.5 %	FW of Kitchen	15

Ref.	Analysis of Ultimate						C/N
	Type of FW	Carbon	Hydrogen	Nitrogen	Oxygen	Sulfur	
[167]		41.5 %	5.8 %	1.6 %	51 %	0.12 %	25.9 %
[157]		39 %	7.32 %	5.7 %	47.7 %	<0.3 %	6.8 %
[171]		45.52 %	6.88 %	2.06 %	25.24 %	NA	22.1 %
[152]		50.7 %	7.6 %	4.4 %	36.9 %	0.4 %	11.5 %
[172]		52.66 %	7.60 %	1.66 %	26.74 %	0.09 %	31.72 %
[165]		48.0 %	6.4 %	2.6 %	37.6 %	0.4 %	18.5 %
[173]		47 %	7.4 %	2.7 %	NA	NA	17.5 %
[174]		48 %	6.4 %	2.6 %	37.6 %	0.4 %	18.5 %
[175]		51.4 %	6.1 %	3.5 %	38.9 %	0.1 %	14.7 %

Table 7

Pre-treatment affects the yield of biocrude from hydrothermal liquefaction.

Use of pretreatment processes	Conditions Process (°C)	Pre-treatment biocrude yield change	Raw material	Results	Pressure (MPa)	Ref.
Supercritical and subcritical hydrothermal melting	390	EFB 37.4, PMF 34.3, and PKS 38.5	PKS, EFB, and PMF	PKS should be completed in 240 min, while PMF and EFB should take 120 min.	25/s	[73]
Low temperature	180	The yield reached its maximum of 58.3 wt% at 240 degrees Celsius, having climbed from 34.5 wt% at 180 degrees Celsius to 56 wt% at 300 degrees Celsius.	Cryptococcus curvatus, sometimes known as yeast, biomass	Elimination of sugar as well as protein	30/seconds	[181]
A weak acid composed of distilled water, formic acid (5 wt%), and sulfuric acid (5 wt%)	<200	At 100 degrees Celsius, there is a 32–40 % increase in yield, but at 200 degrees Celsius, there is a drop in production.	Chlorella vulgaris is a kind of microalgae.	Removal of more than fifty-five % of N	5–15/s	[179]

contains a high amount of water, to be utilised as the reaction medium. The hydrothermal liquefaction method has great potential to produce biocrude with FW containing a lot of water content, such as waste sludge and FW [207]. However, the biocrude delivered via hydrothermal liquefaction requires processing using NiMo and CoMo catalysts to increase the HHV to 45.4–47 MJ/kg. It can also reduce the heteroatom content of Oxygen, nitrogen, and sulfur in the range of 85.1 % to 96.5 %. However, high nitrogen content can significantly influence it, so you can use hydrodenitrogenation to reduce it.

4.7. Effects of food waste composition

The type of raw material used for the hydrothermal liquefaction process dramatically determines the quality of the biocrude. Biocrude is produced from 10 % albumin as raw material through hydrothermal liquefaction treatment at a temperature of 300 °C with a processing time of 2 h. The final result of the biocrude is rich in nitrogen in pure protein [65,208]. Raw materials, such as grains, onions, and microalgae, contain high amounts of protein, allowing for the production of higher levels of biocrude [209–212]. The hydrothermal liquefaction process, utilising raw materials such as lignin, cellulose, and hemicellulose, can yield varied results [189,197]. The hydrothermal liquefaction process, conducted for 30 min at a temperature of 300 °C, using a mixture of rice straw and water hyacinth biomass, can produce 11.6 % hemicellulose [213]. Wood biomass waste containing high levels of lignin can produce lignin groups and 20.9 % lignocellulosic bioethanol, processed through hydrothermal liquefaction for 85 min at 350 °C [214]. Lignin treatment is a crucial parameter for the hydrothermal liquefaction process, enabling the conversion of charcoal into biocrude and increasing the

water-solids ratio in the reaction medium [215–217]. Based on the research results above, the raw materials used differ, so the results of the reported biocrude cannot be directly compared. However, lipid parameters can produce higher levels of biocrude because lipids can be hydrolysed more quickly at temperatures above 280 °C [218,219]. The hydrothermal liquefaction treatment reported in several previous studies indicates that waste butter, sunflower oil, castor oil, corn oil, and soybean oil have promising prospects [160,218,220–222].

The biocrude results from the hydrothermal liquefaction process contain O₂, which is lower by around 10–18 % because it has been converted into CO₂. The water phase is also released from the used biomass waste [223–225]. As much as 21 % of biocrude is made from cellulose through hydrothermal liquefaction at a temperature of 250 °C for ± 1 h. This is because high temperatures can help accelerate hydrolysis [226]. However, at a low temperature of 180 °C, it shows more sensitivity to hemicellulose than at a temperature of 250 °C [227]. Hemicellulose and high cellulose content found in biomass play an essential role in the water phase, allowing to produce more biocrude. This is due to the degradation of hemicellulose, which is cheaper and structured to follow cellulose, and lower crystal formation caused by β1-4 glucosidic bonds. It is also due to the degradation of lignin, which is more difficult to form due to the higher level of lignin and polymerisation in the charcoal residue [189,197]. High lignin content in softwood waste produces less biocrude than in hardwood waste [228].

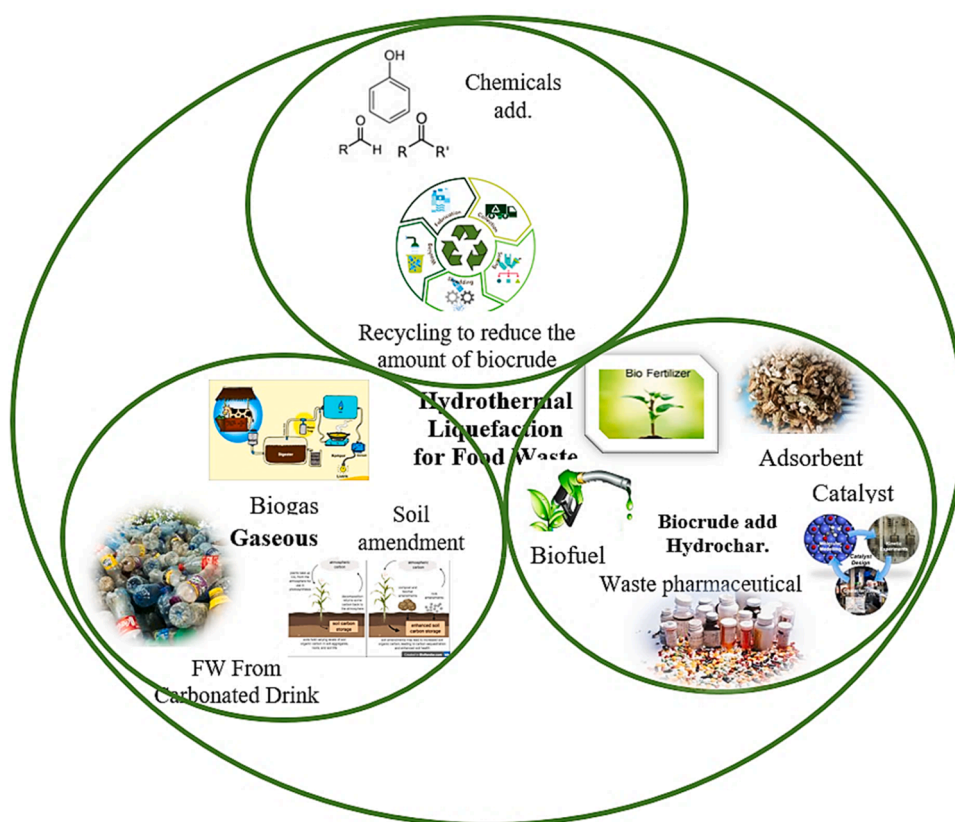
5. Food waste production process and its application

Three product categories are produced from hydrothermal liquefaction processes: water, slurry and gas for biocrude [229–231]. The

Table 8

Biocrude yield and HHV from FW using various catalysts.

HHV (MJ/kg)	Catalyst type (Biocrude yield%)			Temperature operating conditions (°C)	Pressure (MPa)	Type of FW	Ref.
	Using a homogeneous catalyst	Using a heterogeneous catalyst	Without using a catalyst				
NA	NA	Fe-CoZSM-5, 41.8	NA	250	2/h	Soya oil, potato starch, and casein	[199]
Ni/CeO ₂ 34.6.	NA	Ni/	NA	300	20.7/h	A combination of protein and carbohydrates from a variety of sources	[200]
Ni/CeZrOx 37.3.		CeO ₂ 35.6.					
Ni/ZrO ₂ 39.3		Ni/CeZrOx 35.5.					
K ₂ CO ₃ 25.1	NA	Ni/ZrO ₂ 43.2.	NA	300–350	2/s	Peels of the following fruits: watermelon, banana, and pineapple	[201]
21.1–39.4	NA	K ₂ CO ₃ 56.6	NA	250–300	5–20/h	Bovine serum albumin, linoleic acid, and potato starch were the ingredients.	[202]
Acid 32.5.	NA	Catalyst of Alkali	NA	300	2.5/s	leftovers heavy in carbohydrates from the dining hall	[203]
Base 35.4.		27.6–53.8	Base and Acid				
No-additive 34.8		Base 33 and Acid 43					
ZSM-5 34.8.	NA	Base 33 and Acid 43					
ZSM-5 + MgO 32.5		ZSM-5 35.5.	NA	270–310	10–18/s	Peanut shells	[204]
MgO 32.6		ZSM-5 + MgO					
NA	NA	40.6	NA	300	20/h	FW	[205]
K ₂ CO ₃ 34.8	NA	MgO 31.2	NA	320	15/s	Recycle the food scraps from the fields.	[152]
Na ₂ CO ₃ 24.2	Na ₂ CO ₃	Hydroxyapatite 37	Thermal	300	7.6/h	FW for Institutional	[198]
CeZrOx 31.2	21.3	K ₂ CO ₃ 50.3					
Thermal 35.6		CeZrOx 38.8					

**Fig. 7.** Illustration of the application of byproducts generated during hydrothermal liquefaction.

amount of carbon is lower when the gas phase reaches ~7.7 %, the oil phase is high at ~50.3 %, the solid residue is ~24 %, and the rest is the water phase [232]. The results of the hydrothermal liquefaction process with water flow from FW generally contain carbohydrates, isosorbide, acetic acid, oxygenates, ethanol, and acetone [233]. Meanwhile, the number of by-products made from FW raw materials from the hydrothermal liquefaction process ranges from 1 % to 3 % by sample weight. It also contains nitrates, chlorides and sulphates from K and Na [233]. In this case, liquid and solid production during hydrothermal liquefaction can be used as plant fertiliser [234]. Product utilisation is carried out in the development of products with economic value for the community, and as the primary material for hydrothermal liquefaction, as presented in Fig. 7.

5.1. Effect of food waste on greenhouse gas emissions

5.1.1. Global perspective

As much as one-third of the food produced for human consumption worldwide is wasted, amounting to 1.3 billion tons annually. The economic value of this loss is estimated at US\$680 billion in developed countries and US\$310 billion in developing countries. At the same time, around 795 million people in the world suffer from hunger, even though the total FW each year could support up to 2 billion people. FW occurs at every stage of the supply chain, from production to consumption, and is commonly divided into two categories: waste from excessive servings and consumer habits (leftovers), and waste from planning or management errors, whether still edible or not. Both types are environmentally harmful due to their non-recyclable chemical compositions and their contribution to greenhouse gas (GHG) emissions during decomposition. Globally, FW accounts for approximately 8 % of GHG emissions [235], primarily methane, which has a 25 times higher warming potential than CO₂. If FW were considered a country, it would be the third-largest GHG emitter after China and the United States [217].

5.1.2. National case study: Indonesia

In Indonesia, FW has reached an alarming scale, leading to what has been termed a “FW emergency” since 2020. In fact, Indonesia was reported as the second-largest producer of FW globally in 2019, after Saudi Arabia. According to the National Waste Management Information System, FW in 2021 amounted to 46.35 million tons, making it the most significant component of national waste, exceeding even plastic waste (26.27 million tons). The consequences are multidimensional: economically, FW represents a loss of IDR 213–551 trillion annually, while socially, it coincides with persistent malnutrition and stunting problems affecting over 8 million children. Environmentally, Indonesia’s FW generated 1702.9 megatons of CO₂-equivalent emissions over the last two decades, accounting for an average of 7.29 % of annual national GHG emissions [236]. Studies show that one ton of FW produces 4.3 times more GHG emissions than an equivalent amount of edible leftovers, with the consumption stage identified as the most significant contributor [236].

Compared to the global average, where FW accounts for approximately 8 % of GHG emissions [235], Indonesia’s share of 7.29 % [236] indicates that the country contributes nearly at the same level as the worldwide average, despite having a smaller overall population and economic size than major emitters. This comparison highlights the disproportionate severity of Indonesia’s FW problem in relation to its national context, underscoring the urgency of targeted interventions at both policy and household levels.

Table 9 highlights the stark contrast between global and Indonesian FW, underscoring both the scale of the issue and its disproportionate impacts. Globally, one-third of food produced is wasted, amounting to 1.3 billion tons annually, with economic losses reaching US\$680 billion in developed countries and US\$310 billion in developing nations. In comparison, Indonesia alone generated 46.35 million tons of FW in 2021, translating into financial losses of IDR 213–551 trillion each year.

Table 9

Comparison of global and Indonesia FW.

Indicator	Global	Indonesia
Annual FW Volume	1.3 billion tons	46.35 million tons (2021)
Economic Loss from FW	US\$680 billion (developed); US\$310 billion (developing)	IDR 213–551 trillion annually
Population Affected by Hunger	795 million people are hungry	8 million children stunted
Share of GHG Emissions from FW	≈ 8 % of global emissions [217]	≈ 7.29 % of national emissions [218]
GHG Emissions (FW specific)	Equivalent to 3rd largest emitter after China & US [217]	1702.9 Mt CO ₂ -eq over last 20 years [218]

While 795 million people worldwide face hunger, Indonesia simultaneously struggles with 8 million children suffering from stunting, illustrating the paradox of food abundance and malnutrition. From an environmental perspective, FW contributes around 8 % of global GHG emissions [217], making it the third-largest emitter after China and the United States. Indonesia mirrors this trend, with FW accounting for approximately 7.29 % of its national emissions and producing 1702.9 Mt CO₂-eq over the past two decades [218]. This comparison reveals that despite differences in scale, Indonesia’s relative burden is nearly equivalent to the global average, highlighting the urgency of national strategies to reduce FW as part of both climate mitigation and food security efforts.

5.2. Products from food waste

Increasing awareness about environmental issues is directing world attention to alternative foods made from FW. Several countries, including Norway, South Korea, France, and South Africa, were starting to implement FW processing values. This trend is slowly changing people’s assessment of FW. In Indonesia itself, people are beginning to become interested in the issue of FW. However, people still leave food behind and discard it, especially rice and vegetables. Leftover food that is still fit for consumption can be reprocessed into a new menu without sacrificing food quality and nutritional content. Several food products are processed from FW worldwide, as presented in Fig. 8.

Pulp Chips are a snack made from vegetable fibre left over from juice production. The Pulp Pantry company manufactures this product. Kaitlin Mogentale, the founder of Pulp Pantry, was inspired by fibre waste left over from carrot juice production. Seeing the condition of carrot waste being thrown away, he had the idea to process it into something that would benefit the environment and the economy. Pulp Chips has four flavour variants: Jalapeno Lime, Barbecue, Salt and Vinegar, and Sea Salt. Apart from being environmentally friendly, the advantage of Pulp Chips is that they are high in fibre but low in carbohydrates. This product also has several claims that adapt to consumers’ lifestyles [237]. The inventor of the Blue Stripe, Oded Brenner, has been working in the chocolate processing sector for 25 years and is nicknamed the reincarnation of Willy Wonka. He realised that 70 % of the cocoa fruit is wasted during chocolate production. While visiting Jamaica, Brenner was inspired by traditional food made from cocoa beans. Since then, he has viewed chocolate as a superfood, where every part of the fruit can be utilised. Blue Stripe develops several types of products using whole cocoa to utilise all parts of the cocoa. Some products offered are unique, such as Cacao Water and Cacao Smoothies. Both products are processed from cocoa fruit, which is rich in antioxidants [238].

A company from the United States called Fuirtsmart processes the remaining cold-pressed juice production into fibre flour; one of the products is Apple Fibre. This product is produced from Apple Pomace, which consists of apple skin, pulp, and seeds (25 % of the remaining apple production). Apple Fibre can be used in your favourite cooking recipes, such as a chocolate muffin [239]. Apples themselves have twice the antioxidant content of red grapes. Other advantages of apples are



Fig. 8. Food products processed from FW.

that they are low in fat, sodium, and vitamins and are a good source of fibre. Fibre flour is an alternative, environmentally friendly product that still provides health benefits for consumers. Coffee production worldwide is not entirely waste-free, with a byproduct known as cascara. Of the 2 tons of coffee cherries picked, almost 1 ton of cascara, consisting of cherry skin and coffee cherries, is discarded. If reprocessed, cascara tastes like citrus and has high antioxidant content [240]. Processing cascara into tea yields a unique tea product distinct from traditional tea. Its characteristics include low sugar and caffeine. The caffeine level is enough to help improve mood and provide hydration. Additionally, cascara tea is rich in antioxidants and vitamins. It is suitable for consumption by those who pay attention to daily calorie consumption because it contains low calories. Organic Okara Flour is a flour product from upcycled food produced by the Renewal Mill company. The advantage of this product is its high protein content, low carbohydrates, and it is a good source of fibre.

Apart from that, okara flour has a gluten-free claim, which is suitable for consumers who avoid gluten. This flour is made from soybean pulp left over from soybean milk production, thereby helping to reduce FW without sacrificing the quality of the product produced. In line with the Upcycled Food trend, the Surplus Peduli Pangan Foundation regularly provides ideas for processing leftover food into new menus. Some options include pizza made from leftover rice, chicken curry, eggs, steamed banana cupcakes, and many more. The summary of the products above shows that FW still has sales value if managed properly. We can further preserve the environment by managing FW wisely together.

6. Significance

6.1. Social and environmental significance

According to the findings of a recent study, the environmental research revealed that CO₂ emissions can be reduced by three different methods, including the following [241]:

- Due to the existence of renewable fuels as a substitute for fossil energy.
- When compared to the CO₂ emissions projected for the process of converting fossil crude to diesel, the emissions associated with refining and co-refining biocrude obtained from FW are lower.

- Emissions of greenhouse gases are cut down because of diverting organic FW from landfills to be used as a fuel for transportation, which recovers seventy % of its carbon.

It has been estimated that the United States' enormous volume of FW presents an opportunity to reduce annual CO₂ emissions by 15.3 million tonnes. Regarding the societal implications, consuming food that humans or animals waste is unacceptable. As a result, the effective utilisation of FW does not result in the occupation of farmland or an increase in food costs. Since landfills do not contain any FW, diverting the garbage from landfills to be used to manufacture fuel would free up additional space that might be used for cultivation or farming to produce food crops.

6.2. Technological and economic significance

The hydrothermal melting research carried out in a laboratory or on a pilot scale shows promise; however, several other aspects come into play when scaling up to a commercial level. Because the working state requires high pressure and temperature, hydrothermal melting necessitates substantial maintenance to keep the system corrosion-free. This is necessary to maintain the reactor condition and ensure the metered feedstock is being fed properly. As a result, it may be possible to keep the same level of revenue from the processing by employing processing units of a lower scale that are paired with high-pressure pumps of the ideal size [22,23]. It is possible to administer pre-treatments to the feedstock to prevent the need for repeated cleaning and clogging. It designed a mobile pilot-scale reactor with a capacity of 35 litres that operates at a temperature of 300° for 60 min with a feed rate of 0.15 gallons per minute [45]. The pilot plant's continuous reactors produced biocrude with a yield that was 29.5 % higher than that of the lab-scale continuous reactor (21.9 %).

Nevertheless, according to the findings of the techno-economic study, a food distribution network exceeding 160 miles in length has the potential to yield a greater return for the stationary hydrothermal melting reactor. They described the challenges encountered during the installation of a pilot-scale continuous hydrothermal melting, as well as the processing of feedstock and the solutions to those challenges [74]. Compared to, for example, the waste of food on a commercial scale, the economic loss of a country, the depletion of fossil fuels, the growing population, and the need for energy, the above concerns appear to be

relatively insignificant.

Utilising mobile (>170 km), pilot scale (35 L, 0.2 Gal/Min at 300 °C/60 min response time at 1552 psi nitrogen pressure), and lab-size reactors, the technological and economic feasibility of the hydrothermal melting of FW that was obtained from a university dining hall in Champaign, Illinois, was explored in a previous study [45]. The laboratory and the pilot-scale hydrothermal melting tests observed identical physicochemical parameters (ultimate analysis, calorific value, and biocrude composition). We developed three possible outcomes to determine whether the economy would be feasible for producing biocrude using hydrothermal melting: the base case, the best-case scenario, and the worst-case scenario. At the lab scale (30 mL SS tubular reactor) and pilot scale reactors, the hydrothermal melting reactor was operated with a solid content of 13.3 wt% in the feedstock slurry. Dichloromethane was utilised to extract the biocrude from the reactor's output. Using the NRTL thermodynamic model, the system's wastewater treatment and hydrothermal melting procedures were simulated in ASPEN Version 11 (V11). The ASPEN economic estimator V11 was utilised to analyse the overall capital investment. The discount cash flow rate of return approach was used for 30 years to compute the return on assets, and the price at which sales were lowest was determined by utilising the energy density of petroleum as USD/gal of petroleum-equivalent (\$/GGE) values. The cost of utilities, the scale of the reaction, the duration of the project, the yield of biocrude, the price of feedstock, the tax rate, any subsidies, the cost of treating wastewater, and the discount rate are all significant aspects that contributed to the economic analysis of the process. Based on the findings of the financial analysis, it was determined that the wastewater treatment system was more expensive than the biocrude production unit, which accounted for 45 % of the total capital investment. In this investigation, the costs of transporting the feedstock were not considered, and the feedstock prices were assumed to be zero. The scale of the production process greatly impacted the price of the product. Due to the availability of biomass feedstocks, the production of biocrude using woody biomass and algae might be easily scaled up compared to the production of biocrude using other feedstocks. After conducting an in-depth analysis of the production process and financial aspects, it was determined that the minimum selling price (MSP) of biocrude is \$ 3.50 per GGE. The MSP decreased from \$3.5 to \$1.8 per GGE when the biocrude yield was >47 wt% and up to 80 wt%; however, when the biocrude yield is =10 wt%, the MSP is \$16.2 per GGE; for this reason, the biocrude yield is a significant factor in the MSP for an economically feasible hydrothermal melting process. Chemical usage, electricity price, project life, and water treatment would not significantly impact the selling price; nonetheless, government backing makes biofuel manufacturing more competitive in the current market. In conclusion, an on-site hydrothermal melting reactor is ideal for large cities with abundant freshwater, and a transportable unit is suitable for rural locations. As a result, the area of hydrothermal melting systems significantly impacts economic performance; hence, each situation must be scrutinised in minute detail to consider the legislative incentives and on-site constraints.

7. Future perspectives and implications

7.1. Prospects for the future valorisation of food waste

Food loss and waste occur at every stage of the supply chain, from production to consumption. Recycling and bioconversion into bioenergy and other value-added products are effective ways to reduce this problem. Researchers have developed various strategies to recover biofuels and high-value products from FW. Current studies focus on new technologies that maximise output while reducing costs and energy use. Integrated biorefineries offer significant potential, as they enable full utilisation of FW and lower management costs. Several processes, including anaerobic digestion, fermentation, and transesterification, are already applied to produce biomethane, biohydrogen, biodiesel,

bioplastics, and organic acids. Energy-efficient pretreatments and post-treatment technologies can further improve yields and process efficiency. From an economic perspective, integrated FW biorefineries provide the most feasible pathway toward large-scale application. They also support the development of a sustainable circular bioeconomy by repurposing organic waste into valuable bio-based products. Public awareness and action at all stages of the supply chain, from production to consumption, remain crucial to limiting FW generation.

7.2. Policy and practical implications

Implementing policies and practical measures can significantly reduce FW and greenhouse gas emissions, thereby contributing to broader climate change mitigation efforts. Achieving this impact requires collaborative engagement from governments, businesses, and individuals. Based on the findings of this review, effective FW reduction must integrate policy instruments with practical actions, grounded in evidence from existing studies. Governments should establish ambitious, time-bound reduction targets for households and businesses, as demonstrated by national programs that have achieved measurable progress. To further support consumers, standardised food labelling can minimise confusion over expiration dates and prevent premature disposal of edible food, while tax incentives and subsidies can encourage businesses to donate surplus food to food banks, redirecting valuable resources away from landfills and toward vulnerable populations.

At the community and household levels, practical strategies such as meal planning, shopping lists, and proper storage techniques have proven effective in lowering spoilage and over-purchasing. In the food service industry, offering smaller portions reduces plate waste, while food recovery programs divert surplus food from restaurants and grocery stores to those in need. On the technological front, waste diversion schemes (composting, anaerobic digestion) and bioenergy conversion technologies (hydrothermal liquefaction, fermentation) significantly cut methane emissions, aligning with circular economy goals. Collaboration across the food industry, governments, NGOs, and research institutions is essential to optimise supply chains, reduce losses in production and distribution, and scale innovations such as intelligent packaging, predictive analytics, and mobile apps that connect surplus food with charities. Finally, public awareness campaigns and education initiatives remain critical, as consumer behaviour accounts for a significant share of avoidable FW, while investment in research ensures context-specific solutions for different regions.

8. Conclusions

This review aims to synthesise and critically evaluate the current knowledge on the role of FW in greenhouse gas (GHG) emissions and its potential for valorisation into renewable energy and other high-value products. The findings confirm that FW is a significant driver of global warming, contributing approximately 8 % of global emissions and 7.3 % in Indonesia, while also resulting in substantial economic losses and exacerbating food insecurity. At the same time, the evidence demonstrates that conversion and recovery pathways, including composting, anaerobic digestion, fermentation, pyrolysis, and especially hydrothermal liquefaction for high-moisture FW, can substantially mitigate emissions while generating energy, fuels, and bioproducts. By integrating environmental, technological, and policy perspectives, this review highlights FW valorisation as a dual strategy: reducing climate impacts and advancing a circular bioeconomy. Effective mitigation requires coordinated actions at multiple levels, policy instruments such as clear reduction targets, standardised labelling, and food recovery programs; household and community practices like portion control, meal planning, and storage improvements; and industrial measures focused on supply chain optimisation and technological innovation. Overall, the review demonstrates that reducing and valorising FW is not only an environmental necessity but also a practical opportunity to strengthen

energy security, improve resource efficiency, and contribute directly to climate change mitigation.

CRedit authorship contribution statement

Yeggi Darnas: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Emenda Sembiring:** Writing – review & editing, Visualization, Supervision, Methodology, Investigation, Conceptualization. **Benno Rahardyan:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization. **Erdiwansyah:** Writing – original draft, Visualization, Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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